SEQUENTIAL EXECUTION

Line 1: `print("Hello")`

Line 2: `name = input()`

Line 3: `print("Nice...")`

Line 4: `print("Bye!")`

top to bottom.

Each line finishes before the next begins. Like a line of dominoes falling, instructions are triggered in a strict, predictable chain.

INPUT → PROCESS → OUTPUT

INPUT		PROCESS		OUTPUT
Keyboard Files Sensors	→	Calculate Sort Validate	→	Screen Printer Cloud

Example: `age = 2025 - birth_year`

VARIABLES AS STORAGE

Label → Value

name → "Alice"

Values change over time. When `a = a + 1` is executed:

[5] → [6]

The box remains the same, but the content is replaced.

DATA TYPES VISUALIZATION

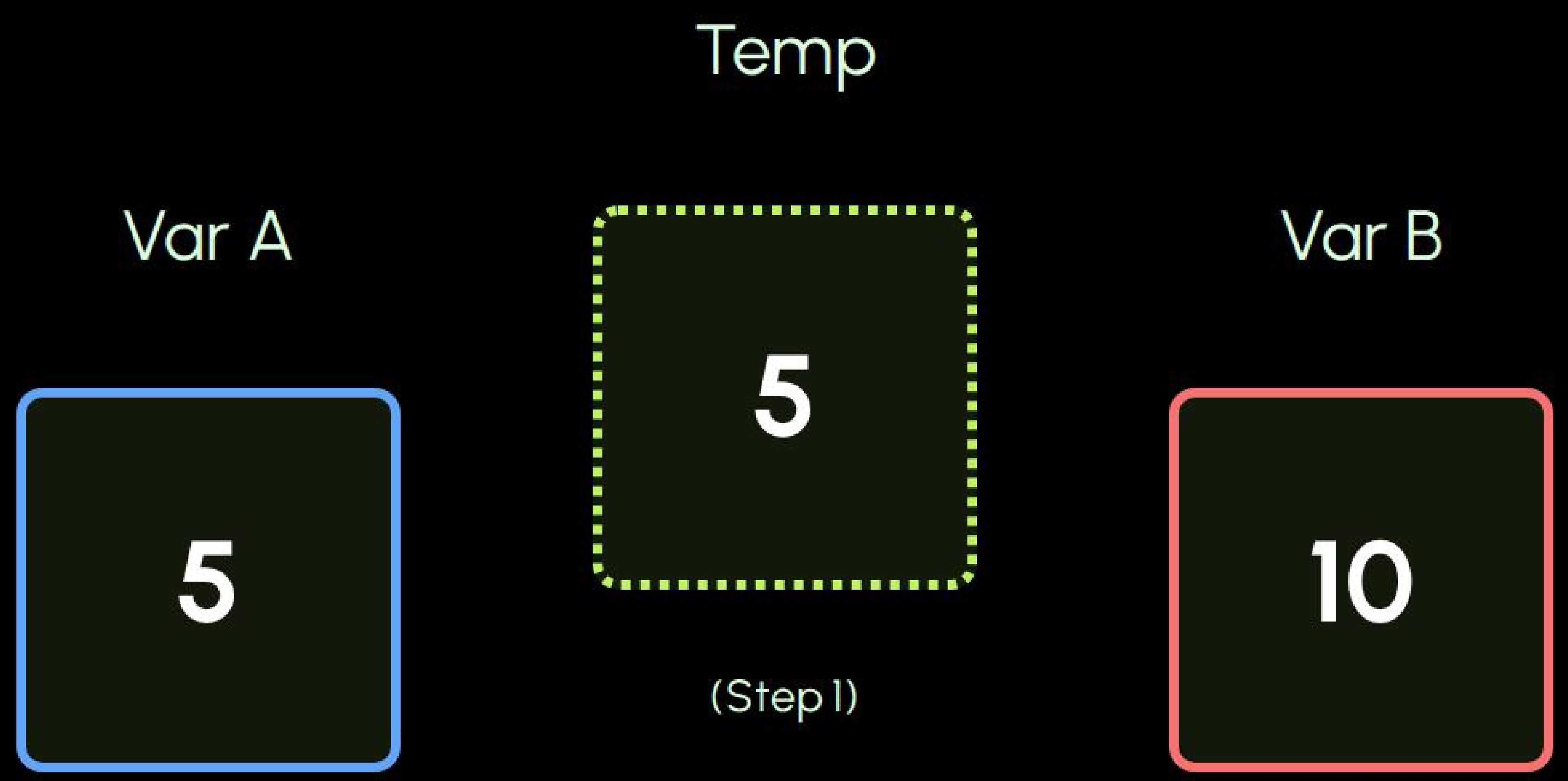
INT / FLOAT
42, 3.14

STRING
"Hello World"

BOOLEAN
True / False

CASTING
`str(42)` → "42"

THE VARIABLE SWAP



1. `temp = a` | 2. `a = b` | 3. `b = temp`

SEQUENTIAL MISTAKES

Undefined Variable

```
 print(score)
score = 10
```

Error: "score" doesn't exist yet in the timeline.

Correct Order

```
 score = 10
print(score)
```

Solution: Define variables before reading them.

Mixed Types

```
 "Age: " + 25
```

Error: Cannot add string and integer.

Type Conversion

```
 "Age: " + str(25)
```

Solution: Cast integer to string first.

EXECUTION: SHOPPING CART

PHASE 1

INPUT

price1 = 10.00
price2 = 15.00
price3 = 20.00

PHASE 2

PROCESS

sub = 10 + 15 + 20 (45)
tax = 45 * 0.08 (3.60)
total = 45 + 3.60

PHASE 3

OUTPUT

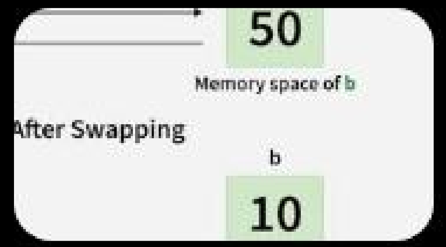
"Total: $48.60"

Trace every value sequentially to avoid logic errors.

IMAGE SOURCES

https://media.istockphoto.com/id/1622410771/vector/infographic-template-5-elements.jpg?s=612x612&w=0&k=20&c=WiUY3JKkypHunL6OSJT70wfxp8zepjNYo9ppAZD3Nfg=

Source: www.istockphoto.com



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